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ORIGINAL CLINICAL RESEARCH REPORT

The Effect of Repeated Bilateral Deep Parasternal Intercostal Plane Catheter Boluses on Analgesia After Cardiac Surgery: A Randomized Controlled Trial

Ronja A. Kuuskoski, MD,[1,2] Teijo I. Saari, MD, PhD,[1,2] Sara I. Karvonen, MS,[3] Bishwesvar Singh, LicSc,[4] Suvi-Maria Tiainen, MD, PhD,[1,5] Matias J. M. Rantanen, MD,[1,2] Minna J. Kallioinen-Sainio, MD, PhD,[1,2] Ville-Veikko Hynninen, MD, PhD,[1,2] Panu I. J. Uusalo, MD, PhD,[1,2] Eliisa Löyttyniemi, MSc,[4] and Marko A. Peltoniemi, MD, PhD[1,2]; Collaborators of TTPcat Study Group

BACKGROUND: Median sternotomy for cardiac surgery is associated with significant postoperative pain. We evaluated whether repeated ropivacaine boluses via bilateral deep parasternal intercostal plane (DPIP) catheters improve analgesia after cardiac surgery.

METHODS: In this randomized, placebo-controlled trial, 120 adult patients undergoing elective coronary artery bypass grafting or heart valve replacement were allocated to receive repeated boluses of either ropivacaine (R) or saline (P) via bilateral DPIP catheters for 72 hours postoperatively. Patients, researchers, and clinical staff were blinded to group allocation. The primary endpoint was postoperative pain intensity, assessed by the visual analog scale or, in sedated patients, the behavioral pain scale. Secondary endpoints included cumulative oxycodone consumption, adverse events, postoperative sedation, and mechanical ventilation.

RESULTS: Pain trajectories over the 72-hour postoperative period did not differ between groups (group × time interaction P = .77 at rest; P = 1.00 with movement). No statistically significant differences were observed between groups in cumulative opioid consumption. For the first 12 hours, the median (95% confidence interval [CI] for median) oxycodone dose was 24 mg (18–27) (P = .55) and from 12 to 24 hours, 24 mg (21–30) in the placebo group and 30 mg (24–39) in the ropivacaine group (P = .32). Corresponding values from 24 to 48 hours were 39 mg (33–45) and 48 mg (39–66) (P = .23), and from 48 to 72 hours 18 mg (12–24) and 24 mg (12–33), respectively (P = .21). Postoperative nausea and vomiting was the most common adverse event, occurring in 38% overall: 18 of 60 (30%) in the placebo group and 28 of 60 (46.7%) in the ropivacaine group, with a higher incidence among ropivacaine-treated valve surgery patients (placebo 9/34 [26.5%] vs ropivacaine 20/33 [60.6%], P = .012). Five patients had symptoms suggestive of local anesthetic systemic toxicity; all cases were self-limiting. Four pneumothoraces occurred, one of which was clearly surgically related. The catheter became dislodged in nine patients. Also, among the dropouts, three patients reported pain during injection of the study drug indicating suboptimal catheter positioning. Postoperative sedation and mechanical ventilation times did not differ between groups. Median (interquartile range and 95% CI for median) time until extubation was 348 minutes (278–412 minutes; CI 337–446 minutes) in the placebo group and 340 minutes (289–539 minutes; CI 400–650 minutes) in the ropivacaine group.

CONCLUSIONS: Repeated ropivacaine boluses via DPIP catheters did not improve postoperative pain control or reduce opioid consumption compared with placebo after median sternotomy. Routine use of DPIP catheters after cardiac surgery should therefore be avoided. Further research is warranted to determine whether continuous infusion techniques or selective use in high-risk patients may yield greater benefit. (Anesth Analg 2026;XXX:00–00)

From the [1]Department of Anesthesiology and Intensive Care, University of Turku, Turku, Finland; [2]Division of Perioperative Services, Intensive Care and Pain Medicine, Turku University Hospital, Turku, Finland; [3]Medical Faculty, University of Turku, Turku, Finland; [4]Department of Biostatistics, University of Turku, Turku, Finland; and [5]Division of Perioperative Services and Intensive Care, Satasairaala Central Hospital, Pori, Finland.

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Data available upon request from the authors. The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Reprints will not be available from the authors.

Address correspondence to Ronja A. Kuuskoski, MD, Division of Perioperative Services, Intensive Care and Pain Medicine, Turku University Hospital, PO Box 51 (Kiinamyllynkatu 4-8), Turku FI-20520, Finland. Address e-mail to ronja.kuuskoski@varha.fi.

KEY POINTS

- **Question:** Does repeated ropivacaine administration via bilateral deep parasternal intercostal plane (DPIP) catheters improve analgesia after median sternotomy when compared with a placebo?
- **Findings:** In this randomized controlled trial of 120 patients, repeated ropivacaine boluses did not reduce pain scores, opioid consumption, or length of stay, and adverse events were similar between groups, with occasional self-limiting symptoms suggestive of local anesthetic toxicity.
- **Meaning:** DPIP catheters with repeated bolus dosing did not provide additional benefit over multimodal analgesia and should not be used routinely after cardiac surgery.

Median sternotomy remains the standard approach for most open cardiac surgeries, with more than two million procedures performed annually worldwide. Postoperative pain following sternotomy is typically severe during the first postoperative days[1] and contributes to delayed recovery and an increased risk of developing chronic pain syndromes in up to 40% of patients.[2,3] Effective pain management is therefore central to enhanced recovery protocols.

Opioids remain the cornerstone of postoperative analgesia in cardiac surgery, but their adverse effects—including respiratory depression, delirium, ileus, and nausea—are of particular concern in this vulnerable population.[4] Neuraxial techniques such as thoracic epidural analgesia provide effective analgesia but are rarely used in cardiac surgery owing to risks of hemodynamic instability and epidural hematoma in anticoagulated patients. These limitations have led to growing interest in ultrasound-guided fascial plane blocks as simpler and safer alternatives.[5]

The deep parasternal intercostal plane (DPIP) block—also referred to in earlier literature as the transversus thoracis plane (TTP) block—targets the anterior branches of intercostal nerves (T2–T6) and provides coverage of the sternotomy wound. A previous randomized controlled trial reported reduced postoperative pain and opioid consumption with single-injection DPIP blocks.[6]. A subsequent randomized controlled trial of continuous DPIP infusion further suggested improvements in pain, opioid use, and recovery parameters.[7] A meta-analysis incorporating these studies concluded that parasternal and DPIP blocks could improve analgesia, but heterogeneity and small sample sizes limited the certainty of evidence.[8]

We designed the present randomized controlled trial to evaluate whether repeated boluses of ropivacaine via bilateral DPIP catheters could provide sustained postoperative analgesia after cardiac surgery. We hypothesized that intermittent dosing might prolong block efficacy compared with single injections, while avoiding the resource requirements of continuous infusion. Our primary outcome was postoperative pain intensity, with secondary outcomes including opioid consumption, adverse events, and length of stay.

METHODS

Ethics and Registration

This study (TTPcat) was approved by the institutional review board of the Hospital District of Southwest Finland (number: T39/2021) and the Finnish Medicines Agency (FIMEA, 14/2021). The trial was registered at clinicaltrials.gov (NCT04916418; Principal investigator: T. I. Saari; date of registration: June 1, 2021) and in the EudraCT database (2020-004178-23, principal investigator: T. I. Saari; date of registration: April 14, 2021) before initiating patient enrollment. No changes to methods were made after trial commencement. The detailed study protocol is available on request from the authors. Written informed consent was obtained from the participants before inclusion in the study.

Patient Population

This study was conducted at Turku University Hospital in Turku, Finland. One hundred and twenty adult patients scheduled for elective open-heart surgery via median sternotomy including coronary arterial bypass grafting (CABG) or single heart valve replacement (HVR) were recruited between October 2021 and February 2024.

Subjects with a history of intolerance to the study drug, alcoholism, substance abuse, or psychological and emotional conditions that could impair their ability to provide informed consent were excluded. Additionally, participants with diagnosed hepatic cirrhosis or severe kidney disease (defined as a glomerular filtration rate <29 mL/min/1.73 m^2 or dependence on dialysis), severe cardiac insufficiency (left ventricular ejection fraction <30%), insulin-dependent diabetes with neuropathy, or a diagnosis of neuropathic pain or chronic pain syndrome were not eligible. Patients with a history of endocarditis or mediastinitis, redo surgery, or combined coronary artery bypass grafting (CABG) and heart valve surgery were also excluded. Concomitant therapy with strong opioids or strong inducers or inhibitors of cytochrome P450 enzymes CYP3A4 or CYP2D6 within 2 weeks before the study was prohibited. Exclusion criteria also included age outside the range of 18 to 80 years, a body mass index >35, body weight <60 kg, diagnosed sleep apnea, or other sleep disorders requiring treatment with a continuous positive airway pressure device. Simultaneous enrollment in other clinical trials, pregnancy, or blood donation within or 4 weeks before study also led to exclusion.

Study Design, Randomization, and Blinding

A randomized, placebo-controlled study protocol was used (Figure 1). Patients, clinical staff, and study researchers were blinded to group allocation. Participants were randomly assigned to one of two groups in a 1:1 allocation ratio. The placebo group received DPIP block with an initial dose of placebo (NaCl 0.9%) 20 mL per catheter (total volume 40 mL), followed by 20 mL per catheter every 8 hours until 72 hours from the first dose. The ropivacaine group received DPIP block with an initial dose of ropivacaine 0.5% (5 mg/mL) 20 mL per catheter (total volume 40 mL), followed by doses of ropivacaine 0.2% (2 mg/mL) 20 mL per catheter every 8 hours until 72 hours from the first dose.

An independent statistician created a computer-generated randomization list using a random permuted block design in randomization with a block size of six (SAS software, Version 9.4 of the SAS System for

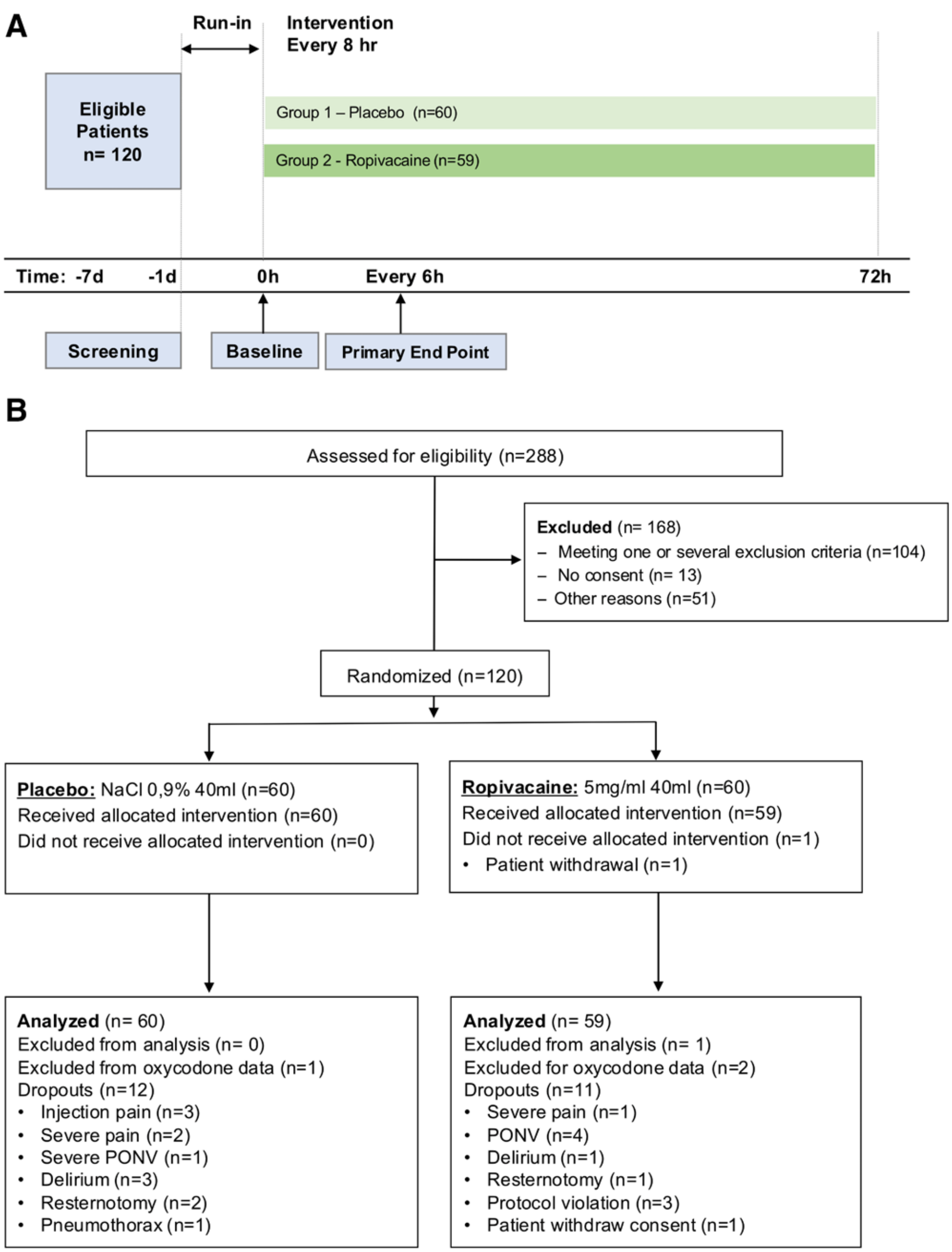

Figure 1. Study design and participant flow of the TTPcat randomized controlled trial. A, Trial flow showing the groups and timeline for the DPIP catheter bolus trial (NCT04916418). B, CONSORT diagram of screening, randomization, allocation, follow-up, and analysis. Numbers shown for exclusions and attrition at each stage. CONSORT indicates CONsolidated Standards of Reporting Trials; DPIP, deep parasternal intercostal plane; PONV, postoperative nausea and vomiting.

Windows, SAS Institute Inc). The ward pharmacist randomized the patients and ordered the study drugs from the local hospital pharmacy, which took care of the assignment. Coded syringes with no other markings were delivered to the operating room by the pharmacy on the day of surgery to ensure the blinding of patients, researchers, and clinical staff to group allocation.

Conduct of the Study, Anesthesia, and TTP Blocks

After providing written informed consent, patients were instructed in using the visual analog scale (VAS; 0 denotes no pain, and 100 indicates the worst imaginable pain) and the patient-controlled analgesia (PCA) system (CADD-Solis VIP; Smiths Medical).

All patients received 20 mg of oral temazepam as premedication 1 to 2 hours before surgery. Routine anesthetic monitoring included pulse oximetry, electrocardiography, invasive blood pressure via radial artery cannulation, and body temperature measurement. The pulmonary artery catheter was used if necessary, following local protocols.

General anesthesia was induced with propofol (Propofol-Lipuro 10 mg/mL, B. Braun Melsungen AG) and sufentanil 0.5 µg/kg (Sufentanil-Hameln). Anesthesia was maintained with sevoflurane (Ultrane sevoflurane; AbbVie) and continuous infusion of sufentanil at 0.3 $\mu g \cdot kg^{-1} \cdot h^{-1}$. Opioids other than sufentanil, as well as ketamine or α-2-agonists, were not used before or during anesthesia.

Rocuronium bromide (Rocuronium B. Braun 10 mg/mL, 0.6–1.0 mg/kg, B. Braun Melsungen AG) was administered during induction to facilitate endotracheal intubation. The depth of anesthesia was continuously monitored with the bispectral index monitor (The Philips BIS module, Philips Medical Systems). Mean arterial pressure was maintained between 65 and 75 mm Hg, or as per hospital routines, during cannulation or decannulation of aorta and during perfusion with the heart-lung machine. If the mean arterial pressure fell below the targeted range, an intravenous bolus of ephedrine or phenylephrine and norepinephrine infusion was initiated if needed. Additional vasoactive drugs were used as required. Total sufentanil consumption was recorded after anesthesia.

Ultrasound-guided bilateral DPIP blocks were performed on cases as described by Chen et al[9] by the cardiac anesthesiologist at the end of the surgery after securing the sternal wound. The injection was done at the T3 to T4 level in immediate proximity to the sternum (within 2 cm). A Pajunk E-cath (Pajunk GmbH) 51 or 75 mm catheter over needle technique was used depending on the patients' anatomy. The needle was placed between the intercostal muscle and the transversus thoracis muscle using an in-plane view. The first 10 mL of placebo or ropivacaine 0.5% was injected via the needle and the next 10 mL via the catheter to ensure proper spreading of the block by ultrasound. The block was termed successful when the study drug was seen spreading in the craniocaudal direction between the fascias reaching under the upper and lower ribs adjacent to the injection site and pressing down the pleura. The catheter placement was on a single level in the DPIP space. The catheters were secured with adhesive tape.

A PCA pump was attached to the intravenous line and activated after the setting of the DPIP catheters. The dose of oxycodone (Oxycodone Orion 10 $mg \cdot mL^{-1}$; Orion Pharma) in the PCA solution was 3 mg with a 10-minute lockout interval. No continuous background infusion was administered during the postoperative phase. After extubation in the intensive care unit (ICU), the patient was encouraged to use the PCA for postoperative pain management as needed. If the patient was evaluated as suffering from pain but was unable to use the PCA device, assistance was provided by the ICU nurse. The total duration of PCA oxycodone treatment was 3 days (72 hours). No oral opioids or ketamine were used. All analgesic treatment, excluding paracetamol, was delivered through the PCA device. The PCA pump recorded all attempted but unsuccessful and all successful administrations of intravenous oxycodone during the 72-hour period. All PCA pump data were collected and transferred to electronic format for evaluation. Each participant was given 1000 mg of paracetamol by intravenous infusion in 15 minutes (Paracetamol 10 mg/mL 100 mL) at the time of emergence and this administration was continued every 8 hours during the study period. Postoperative nausea and vomiting (PONV) was treated with an intravenous 4 mg bolus of ondansetron and 0.625 mg of dehydrobenzperidol if necessary.

During recovery, the patients' pain was assessed using a VAS (0–100). When the patient's clinical status did not permit VAS assessment, pain was assessed using a numerical rating scale (NRS, range 0–100). In the case of VAS 50 or higher, more intensive use of PCA was encouraged. Behavioral Pain Scale (BPS) was used to evaluate pain and oxycodone need in the ICU when the patient was sedated or under mechanical ventilation.

All study participants were admitted to the ICU for the initial postoperative night, followed by transfer to the cardiac care unit or cardiothoracic surgery ward, in accordance with standard hospital protocols for cardiac surgery patients. The patients were monitored according to hospital standards.

Measurements and Data Handling

The following were recorded every 6 hours from the end of surgery until 72 hours: peripheral oxygen saturation, respiratory rate, heart rate, five-lead electrocardiogram, blood pressure, VAS (0–100) for pain intensity at rest and on movement, location of maximal pain, patients pain relief satisfaction (yes/no), any adverse effects related to the study drug, local anesthetic systemic toxicity (LAST) or symptoms such as nausea, pruritus, ear ringing, dizziness, metallic taste, tremors, tongue numbness, slurred speech, convulsions, mediastinitis, hematoma, pneumothorax, or other were recorded every 6 hours from the end of surgery until 72 hours. During ICU stay, arterial oxygen and carbon dioxide levels (kPa), P/F ratio, and level of sedation with the Richmond Agitation Sedation scale (RASS) were recorded. Cumulative amount of oxycodone and the timing of pressings were saved on the PCA pumps. Timing of catheter removal, reason for removal (planned/unplanned), and the potential insertion of a new catheter were marked as were the removal times for surgery-related drains such as pleural or mediastinal drains. Any concomitant drug use was recorded.

All clinical patient data were collected on individual case report forms. All data were subsequently transferred in electronic format to the local REDCap database (https://redcap.utu.fi/; see https://project-redcap.org/) for data analysis.

Primary and Secondary Outcomes

In accordance with Consolidated Standards of Reporting Trials (CONSORT)(items 14 and 21), the primary outcome was prespecified as postoperative pain intensity measured repeatedly over 72 hours, with inference based on a single group × time interaction from a linear mixed-effects model. No single time point was designated as the primary comparison. Secondary outcomes, including cumulative opioid consumption and adverse events, were prespecified and analyzed descriptively or exploratory without multiplicity adjustment.

Sample Size Calculation

Our aim was to demonstrate significant differences in patient-reported pain ratings measured with a VAS. In a previous study, the VAS score was 37 (standard deviation [SD] 54%) on the first postoperative day. Based on a two-sample t test power calculation, 52 patients per group and a total of 104 patients will be necessary to demonstrate a 30% reduction from 37 to 26 in VAS ratings (a clinically significant difference) using a level of significance of $P = .05$ and power of 80%. We randomized 120 patients in total (60 per group) allowing 16 dropouts.

Statistical Analyses

Continuous variables are summarized with means and SDs if normally distributed, with median together with lower (Q1) and upper (Q3) quartiles or range otherwise. Standardized differences between the groups were also calculated (to be interpreted |standardized difference| <0.2 to small and |standardized difference| <0.5 as medium size). Pain scores measured at rest and during movement over 72 hours (6–8, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, and 72 hours) were compared between the groups with a linear mixed-effects model suitable for repeated measurements, using compound symmetry covariance structure and Kenward-Roger correction for degrees of freedom. The model included the group (between factor) and time (within-factor) and their interaction (which evaluates whether the mean changes differ between the groups over time). The primary comparison was the group × time interaction, reflecting differences in pain trajectories between groups over time. Model-based estimates with standard errors were used for inference. Descriptive mean (SD) values at prespecified postoperative time points are provided for reference. Other outcomes were prespecified secondary or exploratory endpoints. No formal adjustment for multiple hypothesis testing was applied, as the study was powered for a single primary outcome and secondary and subgroup analyses were considered exploratory and interpreted accordingly. In addition, slice effect was calculated, that is, group differences were tested at each timepoint.

The study was designed to detect differences between groups and was not planned as an equivalence or noninferiority trial; therefore, formal equivalence margins were not prespecified. To support interpretation of nonsignificant findings, effect size estimates, standardized differences, and 95% confidence intervals (CIs) are reported for the primary and secondary outcomes.

Analyses were also performed by surgery type. Location of pain and satisfaction of pain management were compared between the groups by time point using Fisher exact test. The cumulative amount of opioid needed was compared between the groups at each time point with the Kruskal–Wallis test. The data analysis for this article was generated using SAS software, Version 9.4 of the SAS System for Windows (SAS Institute Inc).

RESULTS

Participants

A total of 552 elective CABG or HVR patients were operated within the study period. Of these patients, 288 were screened for eligibility and 120 patients were recruited between October 2021 and January 2024 (Figure 1).

Altogether, 120 patients were included in the study. One patient withdrew consent after randomization resulting in missing outcome data. A total of 60 (placebo) and 60 (ropivacaine) patients were assessed for group demographics and 59 (placebo) and 60 (ropivacaine) for other statistical analyses. Due to technical difficulties with the PCA pump in two patients, oxycodone consumption data were unavailable, leading to their exclusion from the oxycodone consumption analysis. Oxycodone consumption was analyzed in 59 patients in the placebo group and 58 patients in the ropivacaine group.

During the study, a total of 23 (placebo n = 12/ ropivacaine n = 11) dropouts occurred. Three patients withdrew from the study due to pain associated with the study drug injection (3/0), three patients due to inadequate pain control (2/1), five patients due to severe PONV (1/4), and one patient did not specify a reason (0/1). The study was discontinued prematurely in eleven patients at the discretion of the investigators due to delirium (n = 4, 3/1), resternotomy (n = 3, 2/1), protocol violation (n = 3, 0/3), and one pneumothorax (1/0). For these patients, data were analyzed until the time of study withdrawal. Four of these dropouts occurred within 24 hours, five within 24 to 36 hours, four within 36 to 48 hours, and 10 within 48 to 72 hours. The catheter became dislodged and was subsequently reinserted in nine patients.

Characteristics of the Study Population

The median age of the patients was 66 (25–79) and 82.5% were men. CABG was performed in 44% of patients, and HVR in 56%. No statistically significant differences were observed between the groups in the examined variables (Table 1). All patients were Caucasian.

Primary Outcome Evaluation

No significant overall group mean differences in pain scores were observed at rest; *P* values for group differences and group × time interaction were .07 and .77, respectively; during movement, the corresponding *P* values were .23 and 1.00. There were no statistically significant differences in mean pain score trajectories between the placebo and ropivacaine groups over the study period. The removal of surgical drains was associated with reduced pain scores in both groups. No statistically significant between-group differences were observed in pain reduction following drain removal (*P* = .88 at rest; *P* = .33 during movement). Detailed results are presented in Table 2 and Supplemental Digital Content S3, Supplemental Table 1, https://links.lww.com/AA/F886 and Figure 2.

Pain scores did not differ significantly between groups when analyzed separately by surgery type (CABG or HVR; Supplemental Digital Contents S1 and S2, Supplemental Figures 1 and 2, https://links.lww.com/AA/F886, Supplemental Digital Contents S4 and S5, Supplemental Tables 2 and 3, https://links.lww.com/AA/F886). No significant mean differences were observed between surgery types (all *P* values >.21).

Table 1. Perioperative Data and Characteristics of 120 Sternotomy Patients Included in the Analysis

	All (n = 120)	Placebo group (n = 60)	Ropivacaine group (n = 60)	Standardized differences <0.2 (small) <0.5 (medium)
Age (y), median, (range)	66 [25–79]	66 [38–78]	65 [25–79]	−0.21
Female/male, n (%)	21/99 (17.5%)[a]	8/52 (13.3%)[a]	13/47 (21.7%)[a]	0.21
Body weight (kg)	82.9 (13.0)	83.6 (13.6)	82.2 (12.4)	−0.09
BMI (kg/m^{-2})	26.7 (3.7)	27.0 (3.6)	26.5 (3.8)	−0.12
Smoking (yes/no), n (%)	18/102 (15.0%)[a]	9/51 (15.0%)[a]	9/51 (15.0%)[a]	−0.01
Diabetes (yes/no), n (%)	20/98 (17.0%)[a]	12/47 (20.3%)[a]	8/51 (13.6%)[a]	−0.18
Atrial fibrillation (yes/no), n (%)	19/101 (15.8%)[a]	11/49 (18.3%)[a]	8/52 (13.3%)[a]	−0.14
ASO (yes/no), n (%)	4/116 (3.3%)[a]	1/59 (1.7%)[a]	3/57 (5%)[a]	0.19
Chronic lung disease (yes/no), n (%)	5/115 (4.2%)[a]	2/58 (3.3%)[a]	3/57 (5.0%)[a]	0.08
Neurological disease (yes/no), n (%)	4/115 (3.4)[a]	0/59 (0.0%)[a]	4/56 (6.7%)[a]	0.38
Type of surgery (CABG/heart valve operation), n (%)	52/66 (44%)[a]	26/34 (43%)[a]	26/32 (45%)[a]	0.03
ASA physical status, n (%)				0.01
II	2	0 (0.00)	2 (1.68)	
III	79	41 (34.5)	38 (31.9)	
IV	38	19 (16.0)	19 (16.0)	
Duration of surgery (min) [Q1, Q3]	219 (50.4)	214 (48.8) [182, 233]	224 (51.9) [190, 253]	0.20
Propofol during surgery and ICU (mg·kg^{-1})	20.0 (12.7)	18.6 (8.6)	21.3 (15.7)	0.22
Sufentanil during surgery (µg·kg^{-1})	1.76 (0.32)	1.74 (0.29)	1.77 (0.32)	0.10
Noradrenaline during ICU (mg·kg^{-1})	0.08 (0.1)	0.06 (0.07)	0.09 (0.12)	0.28

Data are mean (standard deviation) or [quartiles] unless otherwise stated. Standardized difference is the difference in the mean of a variable between the two intervention groups divided by an estimate of the standard deviation of that variable.

Abbreviations: ASA, American Society of Anesthesiologists; ASO, arteriosclerosis obliterans; BMI, body mass index; CABG, coronary artery bypass grafting; ICU, intensive care unit; n, number of non-missing values.

[a]Percentage of females or yes-answers.

Table 2. Pain at Rest and in Movement Measured With VAS at Predefined Timepoints in Placebo and Ropivacaine Groups in 119 Sternotomy Patients Included in the DPIP Catheter Bolus Trial and the Effect of Drain Removal on Pain; Model-Based Means and CIs

	VAS in rest				VAS in movement			
Time after the first dose	All (n = 119)	Placebo (n = 60)	Ropivacaine (n = 59)	*P* value .77	All (n = 119)	Placebo (n = 60)	Ropivacaine (n = 59)	*P* value 1.00
6–8 h								
Mean	42.2	40.0	44.3		50.4	49.7	51.2	
CI	37.5–46.8	33.4–46.6	37.7–51.0		44.7–56.2	41.7–57.7	42.9–59.4	
12 h								
Mean	34.9	34.9	35.0		47.7	46.6	48.9	
CI	30.9–38.9	29.3–40.5	29.2–40.7		42.9–52.5	40.0–53.1	41.9–55.8	
18 h								
Mean	30.1	28.0	32.2		46.8	44.3	49.4	
CI	26.2–34.1	22.5–33.6	26.6–37.8		42.2–51.5	37.9–50.7	42.6–56.1	
24 h								
Mean	30.0	28.4	31.6		50.1	48.8	51.5	
CI	26.0–33.9	22.9–33.8	26.0–37.2		45.5–54.8	42.3–55.3	44.9–58.1	
48 h								
Mean	22.7	19.5	25.9		39.3	39.0	39.7	
CI	18.7–26.7	13.9–25.2	20.2–31.6		34.5–44.1	32.1–45.8	33.0–46.4	
72 h								
Mean	14.8	12.6	16.9		29.5	26.2	32.8	
CI	10.4–19.1	6.5–18.7	10.7–23.2		24.3–34.6	18.9–33.4	25.1–40.1	
Drain removal								
Before removal				*P* value				*P* value
Mean	25.8	24.7	26.9	.88	43.4	44.6	42.1	.33
CI	21.9–29.6	19.4–30.0	21.4–32.4		38.8–48.0	38.3–51.1	35.4–48.7	
After removal								
Mean	20.2	19.5	21.0		38.3	37.0	39.7	
CI	16.0–24.5	13.5–25.5	14.9–27.1		33.1–43.6	29.7–44.3	32.2–47.1	

Reported *P* value in tables is group by time interaction.
Abbreviations: CI, confidence interval; DPIP, deep parasternal intercostal plane; VAS, visual analog scale.

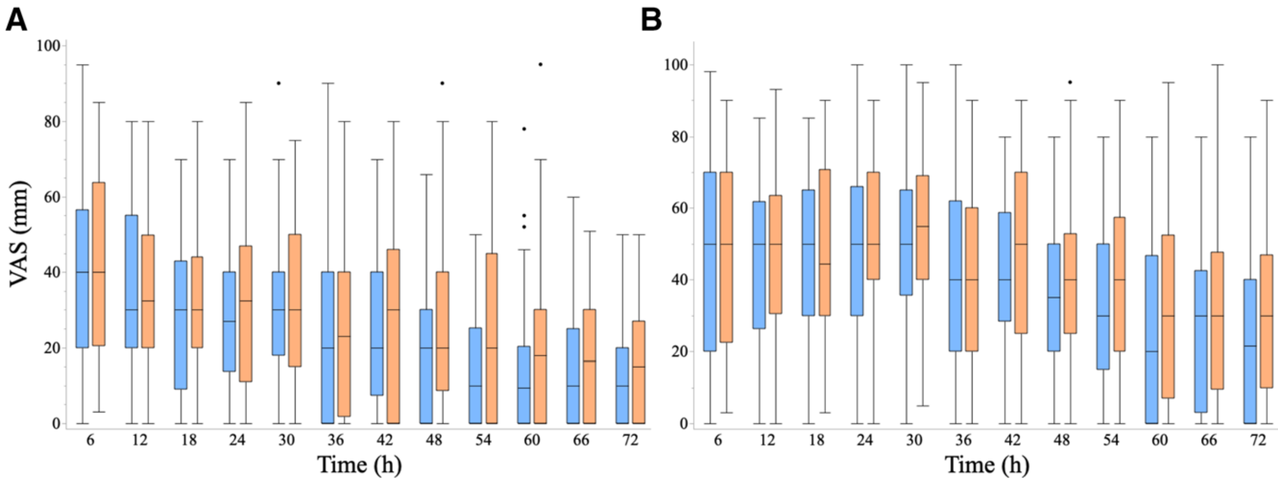

Figure 2. Pain trajectories over 72 h in placebo and ropivacaine groups in 119 sternotomy patients recruited to DPIP catheter bolus trial. Median (line) and interquartile range (shaded band) for VAS at rest (A) and with movement (B) from 0 to 72 h after surgery, by treatment group (placebo versus ropivacaine DPIP). Dotted line marks VAS = 30 (moderate pain threshold). DPIP indicates deep parasternal intercostal plane; VAS, visual analog scale.

Opioid Consumption

No statistically significant differences were observed between groups in cumulative opioid consumption (at 0–8, 0–12, 0–24, 0–48, or 0–72 hours); all *P* values were >.21. For the first 12 hours, the median (95% CI for median) oxycodone dose was 24 mg (18–27) (*P* = .55) and from 12 to 24 hours, 24 mg (21–30) in the placebo group and 30 mg (24–39) in the ropivacaine group (*P* = .32). Corresponding values from 24 to 48 hours were 39 mg (33–45) and 48 mg (39–66) (*P* = .23), and from 48 to 72 hours, 18 mg (12–24) and 24 mg (12–33), respectively (*P* = .21). Detailed results are presented in Table 3 and Figure 3.

The timing of the first opioid dose after extubation did not differ significantly between groups. Surgery type (CABG or HVR) had no significant effect on opioid consumption (all *P* values >.14; Supplemental Digital Content S6, Supplemental Table 4, https://links.lww.com/AA/F886). Removal of surgical drains was associated with reduced oxycodone consumption in both groups (*P* = .48) (Supplemental Digital Content S7, Supplemental Table 5, https://links.lww.com/AA/F886).

Adverse Effects and LAST

The overall incidence of adverse events did not differ significantly between the placebo and ropivacaine groups. PONV was the most frequently reported adverse event (P 28/60 [46.7%] vs R 18/60 [30%], *P* = .09). Subgroup analysis revealed a higher incidence (26.5% vs 60.6%) of PONV in the ropivacaine group undergoing HVR (9/34 and 20/33, in placebo and ropivacaine groups, respectively, *P* = .01).

Other reported adverse effects included dizziness, tremors, muscle cramps, metallic taste, tinnitus, tongue numbness, pruritus, and pneumothorax. These occurred as isolated cases, precluding statistical comparison. Four pneumothoraces were observed during the study period, distributed across both groups. One pneumothorax was clearly surgery-related and detected before study intervention; three may have been related to the DPIP block procedure, including one identified at the end of the study period. No cases of mediastinitis, catheter-site infection, or convulsions were recorded.

Five patients in the ropivacaine group and none in the placebo group reported symptoms potentially compatible with LAST, including tremors, PONV, metallic taste, slurred speech, tinnitus, and dizziness. These symptoms were not reported to study investigators at the time of occurrence, limiting formal assessment of their relationship to local anesthetic exposure. All symptoms resolved spontaneously without medical intervention and resulted in no lasting sequelae.

Table 3. Oxycodone Consumption as i.v. Morphine Equivalents (mg) in 116 Sternotomy Patients Included in the DPIP Catheter Bolus Trial[a]

	Placebo group n = 59	Ropivacaine group n = 57
0–12 h		
Median [Q1, Q3]	24.0 [15.0, 30.0]	21.0 [15.0, 30.0]
CI for median	18.0–27.0	18.0–27.0
12–24 h		
Median [Q1, Q3]	24 [15.0, 36.0]	30.0 [18.0, 48.0]
CI for median	21.0–30.0	24.0–39.0
24–48 h		
Median [Q1, Q3]	39.0 [30.0, 57.0]	48.0 [30.0, 72.0]
CI for median	33.0–45.0	39.0–66.0
48–72 h		
Median [Q1, Q3]	18 [6.0, 36.0]	24.0 [3.0, 38.2]
CI for median	12.0–24.0	12.0–33.0

Opioid consumption is calculated from initiation of the study intervention (first ropivacaine/placebo bolus).

Abbreviations: CI, confidence interval; DPIP, deep parasternal intercostal plane.

[a]There were no differences between the groups.

ICU and Hospital Length of Stay

There were no significant differences between groups in either the ICU or hospital length of stay. The median time (interquartile range) for the ICU period in the placebo group was 23.2 hours (22.2–25.5 hours)

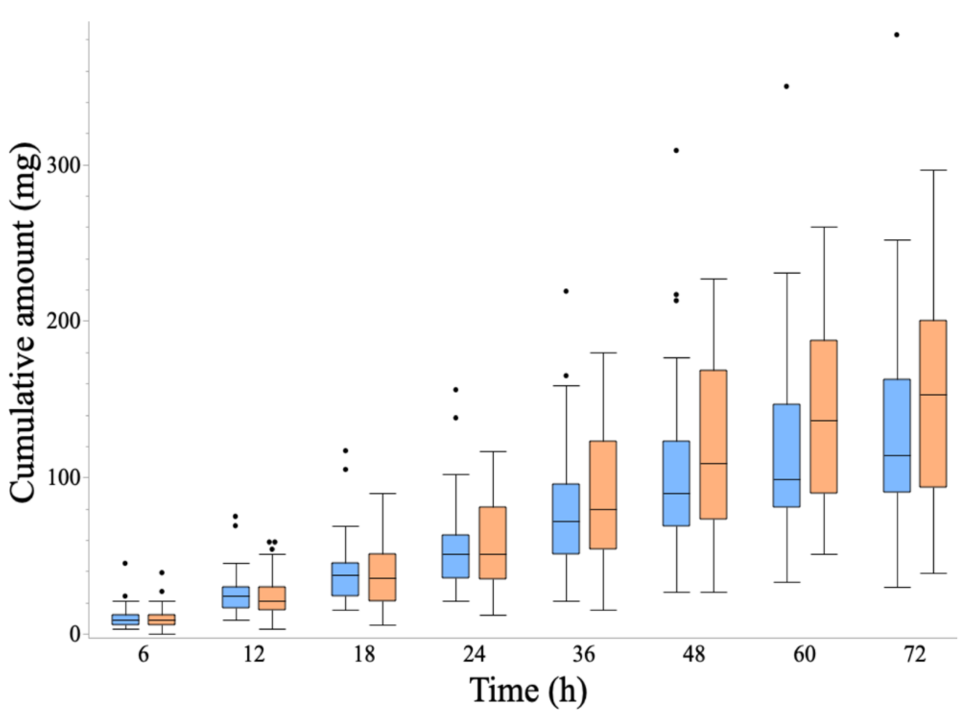

Figure 3. Cumulative oxycodone consumption (in mg) during 72 h in placebo and ropivacaine groups in 116 sternotomy patients recruited to deep parasternal intercostal plane (DPIP) catheter bolus trial.

and 23.1 hours (21.8–25.3 hours) in the ropivacaine group (P = .96). Hospital length of stay for the placebo group was 168 hours (146–172 hours) and 170 hours (145–190 hours) for the ropivacaine group (P = .42).

Postoperative Sedation and Mechanical Ventilation

Postoperative sedation was assessed using the RASS. No statistically significant differences were observed between the placebo and ropivacaine groups in terms of RASS scores or tolerance to mechanical ventilation.

At the 6- to 8-hour assessment, most patients (n = 63, 53%) had been successfully weaned from sedation. By the 12-hour time point, 49 patients in the placebo and 41 patients in the ropivacaine groups had a RASS score of 0. At the same time point, five patients in the placebo and seven patients in the ropivacaine group remained on mechanical ventilation. There were no significant differences in mechanical ventilation between the groups (P > .06).

Median (interquartile range and 95% CI for median) time until extubation was 348 minutes (278–412 minutes; CI, 337–446 minutes) in the placebo group and 340 minutes (289–539 minutes; CI, 400–650 minutes) in the ropivacaine group, with no statistically significant difference between the groups (P = .39).

DISCUSSION

Our findings demonstrate that repeated boluses of ropivacaine via bilateral DPIP (also termed TTP in earlier reports) catheters placed at the end of surgery did not improve pain scores, opioid consumption, or recovery endpoints compared with placebo. These results contrast with earlier randomized trials, including one evaluating continuous DPIP infusion that reported reductions in pain, opioid consumption, and length of stay[7] and another trial of superficial parasternal intercostal plane catheter boluses suggesting improved analgesia.[10] Although this trial was not designed to formally demonstrate equivalence, the small effect sizes and narrow CIs around the primary outcome suggest that any potential analgesic benefit of repeated DPIP catheter boluses is unlikely to be clinically meaningful.

Our findings closely parallel those of Demarquette et al,[11] who conducted a large randomized controlled trial evaluating both superficial and DPIP blocks after sternotomy.[9] In contrast to earlier smaller trials, Demarquette et al[11] incorporated patient-centered functional outcomes, including quality of recovery (QoR-15), alongside pain scores and opioid consumption, and demonstrated no benefit of either block technique compared with contemporary multimodal analgesia. Importantly, they observed a higher incidence of pneumothorax in the deep block group, raising safety concerns without corresponding clinical benefit.

While our study focused primarily on pain trajectories and opioid use rather than global recovery metrics, the consistency of neutral findings across both trials—despite differences in block technique, dosing strategy, and outcome selection—strengthens the conclusion that parasternal fascial plane blocks provide limited incremental value after median sternotomy. Together, these trials suggest that even when analgesic efficacy is assessed across broader functional domains, parasternal blocks fail to meaningfully improve postoperative recovery.

Several factors may explain the discrepancy between early positive reports and recent negative trials. First, baseline pain burden after sternotomy may be lower than anticipated in modern practice; in both our study and Demarquette et al's,[11] median early postoperative pain scores in controls were around 2 (NRS 0–10), limiting the scope for measurable improvement.[11] Second, the duration of effect of single-injection or intermittent bolus fascial plane blocks may be insufficient to alter recovery trajectories, particularly compared with continuous infusions. Third, sternotomy pain is multifactorial, often arising from rib retraction, drain sites, and wider tissue trauma not fully covered by DPIP or superficial parasternal intercostal blocks. Finally, differences in block timing, technique, and study populations may have contributed to heterogeneity in earlier trials.

We placed the DPIP catheters at the end of surgery, consistent with our hospital's routine practice, whereas previous studies inserted catheters preoperatively. All anesthesiologists performing the blocks were experienced, and the technique had been in routine use at our institution for several years (Supplemental Video S8). Although drug spread was verified by ultrasound, surgical tissue disruption particularly in CABG patients after left internal thoracic artery harvesting may have reduced the visibility of the target plane and compromised efficacy. However, subgroup analyses revealed no difference between CABG and valve patients, suggesting that the block was generally performed correctly. The discrepancy with Zhan et al[7] reporting benefit in valve patients only may reflect other factors, such as differences in surgical trauma or block timing.

Another possible explanation for limited efficacy is catheter stability. During the study period, catheter dislodgement occurred in nine patients, necessitating reinsertion. Additionally, three patients in the placebo group reported pain during bolus administration through the catheter. This was considered most likely attributable to suboptimal catheter positioning or partial catheter dislodgement. Reinsertion of the catheters was offered; however, these patients elected to withdraw from the study. The DPIP space is a thin, artificially created plane, into which catheters can be advanced only a few centimeters. Mobilization after surgery may increase the risk of displacement, and intermittent bolus administration may not keep the space "open" as effectively as continuous infusion. This could reduce the durability of analgesia compared with continuous techniques.

In addition to efficacy, the practical feasibility of the technique deserves consideration. During the study, the nine catheters requiring reinsertion were replaced either by the on-call cardiac anesthesiologist or by the principal investigator and could typically be performed at the bedside in the ICU or cardiac care unit where ultrasound equipment was readily available. Although catheter replacement was feasible and not associated with major complications, it required the availability of anesthesiologists experienced in this technique. Such expertise may not always be readily available, particularly outside regular working hours. From a practical perspective, the need for catheter replacement in approximately 7% to 8% of patients represents a non-negligible resource burden that should be considered when evaluating the clinical utility of DPIP catheter techniques.

Catheter instability may partly reflect the anatomical characteristics of the DPIP plane itself. Because the target space is relatively narrow and often created artificially during the block, only limited catheter advancement is possible and postoperative mobilization may predispose to displacement. These factors could reduce catheter stability and contribute to variability in analgesic efficacy. A cadaver study by Douglas et al[12] discovered that the internal mammary artery may be close to DPIP, and extensive skill sets are required to achieve this block safely. Nevertheless, catheter displacement alone is unlikely to fully explain the lack of analgesic benefit observed in our trial, as the absence of effect was evident already in the early postoperative period and is consistent with several recent randomized trials evaluating parasternal block techniques.

Finally, the DPIP block primarily anesthetizes anterior intercostal nerves (T2–T6) and does not cover all nociceptive sources after sternotomy. Pain from rib retraction, drain sites, and wider tissue trauma may remain untreated. In our trial, placebo-treated patients more often localized pain to the sternum, whereas ropivacaine-treated patients reported drain-site discomfort at 24 hours. Although inconsistent across time points and sometimes difficult for patients to distinguish, this finding underscores that sternotomy pain is multifactorial and only partially addressed by DPIP blockade. Limited dermatomal coverage may therefore have contributed to the lack of overall benefit.

The DPIP space lies deep in the thoracic wall, close to the pleura, increasing the risk of complications such as pneumothorax compared with more superficial blocks. From a safety perspective, our study identified only a few possible ropivacaine-related adverse effects and a small number of pneumothoraxes, mostly surgical in origin. Demarquette et al[11] nonetheless reported pneumothorax as a significant complication of deep parasternal blocks, raising the question of whether potential risks are justified when the analgesic benefit appears negligible.

Study Limitations

This study has several limitations. First, it was conducted at a single tertiary center, which may limit generalizability despite standardized protocols and experienced operators. Second, the blocks were performed after sternal closure; thus, tissue disruption from surgery might have hindered the spread of local anesthetic compared with preoperative placement. Third, although repeated boluses were chosen to maintain block patency, intermittent dosing may not replicate the pharmacodynamic stability of continuous infusions, which showed benefit in earlier trials. Fourth, sternotomy pain is multifactorial and not fully addressed by DPIP blockade, particularly at drain sites. Fifth, the multimodal analgesia regimen may differ between countries, and this may affect the generalizability of our study. Sixth, despite rigorous blinding, variability in catheter positioning and postoperative mobilization may have influenced block effectiveness. Finally, the number of dropouts exceeded expectations and may have influenced our results. Most dropouts occurred after the first 24 hours and were primarily related to inadequate pain control or PONV. Only four dropouts occurred within the first 24 hours, supporting the robustness of early postoperative findings; however, differential dropout related to pain or adverse effects may have introduced bias in later time periods and should be considered when interpreting the results.

CONCLUSIONS

Overall, our trial and the recent large negative randomized controlled trial (RCT) suggest that routine use of parasternal or DPIP blocks after cardiac surgery should be avoided. Continuous infusion techniques may hold promise, but their clinical utility requires validation in larger multicenter trials. Future studies should identify patient subgroups at the highest risk of severe acute or chronic post-sternotomy pain, in whom regional anesthesia might provide greater benefit, and compare continuous catheter techniques with more superficial fascial plane approaches.

DISCLOSURES

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ASSOCIATED VIDEO

Video. Performing the DPIP block with ultrasound imaging. The video can be accessed at https://journals.lww.com/cases-anesthesia-analgesia/10.1213/ANE.0000000000008211

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